

Lab Practical Task 01 — RC Servo-Controlled Gate Mechanism (ATmega32) [30 Marks]

Objective:

1. To learn PWM and how to control an RC servo motor.
2. Design and implement a gate-control mechanism using a **servo motor** driven by **Timer1 PWM** on the **ATmega32**. The system must use an external interrupt to detect a button press and perform a timed open/close sequence.

System Requirements

1. **Initial Condition**
 - The gate is **closed** at startup.
2. **Trigger**
 - A **button press** (wired to an external interrupt pin) triggers the sequence.
3. **Gate Opening**
 - On button press, the gate must **open smoothly to 90°** (maximum opening angle).
 - Use **Timer1 PWM** (configured for standard servo signal, e.g. 50 Hz) to control the servo.
4. **Open Delay**
 - The gate remains fully open for **10 seconds**.
5. **Controlled Closing Sequence**
 - After the 10-second delay, the gate must **close in three stages**:
 - **Stage 1:** Close **40°**, then **delay 500 ms**.
 - **Stage 2:** Close the next **30°**, then **delay 500 ms**.
 - **Stage 3:** Close the final **20°** (completing the 90° closure).

Each stage must be implemented as a smooth motion (incremental PWM updates) so that movement is visible and not abrupt.

